

Android App based Exploratory Test Automation

samsung

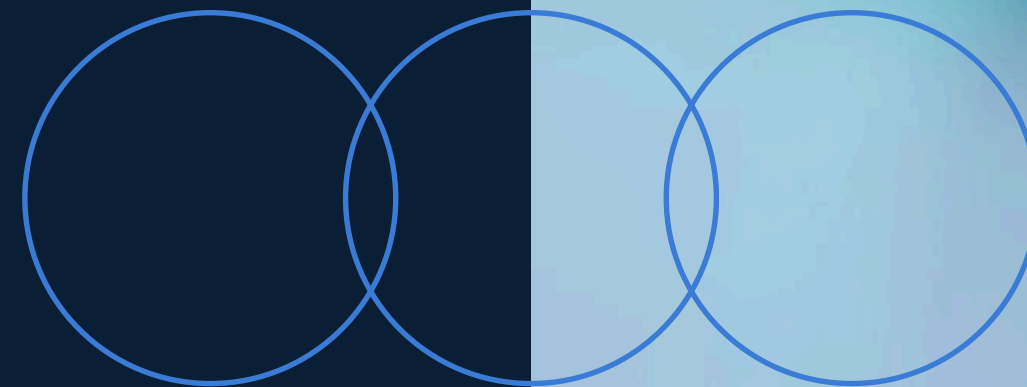
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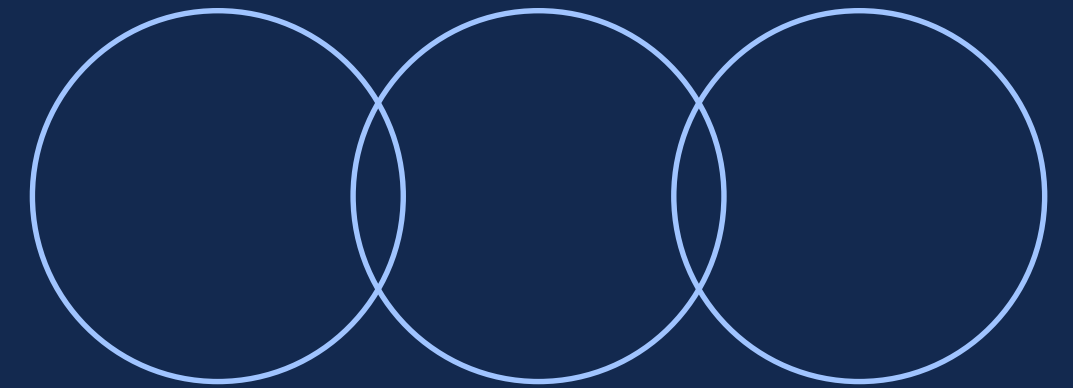


Testing Types

Functional Testing : *Verifies that the software works according to specified requirements
Test cases are designed in advance*

- Predefined test cases
- Structured & planned
- Tests known requirements

more on that later!!!

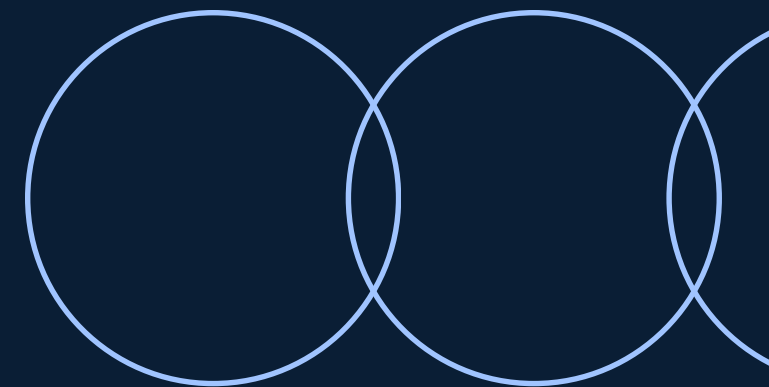


Exploratory Testing : *Discovers unknown defects by actively exploring the system, Finding edge cases, usability issues, hidden bugs*

- Based on tester's knowledge, experience, and intuition
- But this is Mostly **manual**, human-driven
- Tests known requirements

Why is exploratory testing difficult?

- Mostly **manual and human-driven**
- Requires strong domain knowledge
- Hard to reproduce and document
- Inconsistent coverage across testers
- Critical bugs may be missed under tight deadlines
- No specific benchmarks.

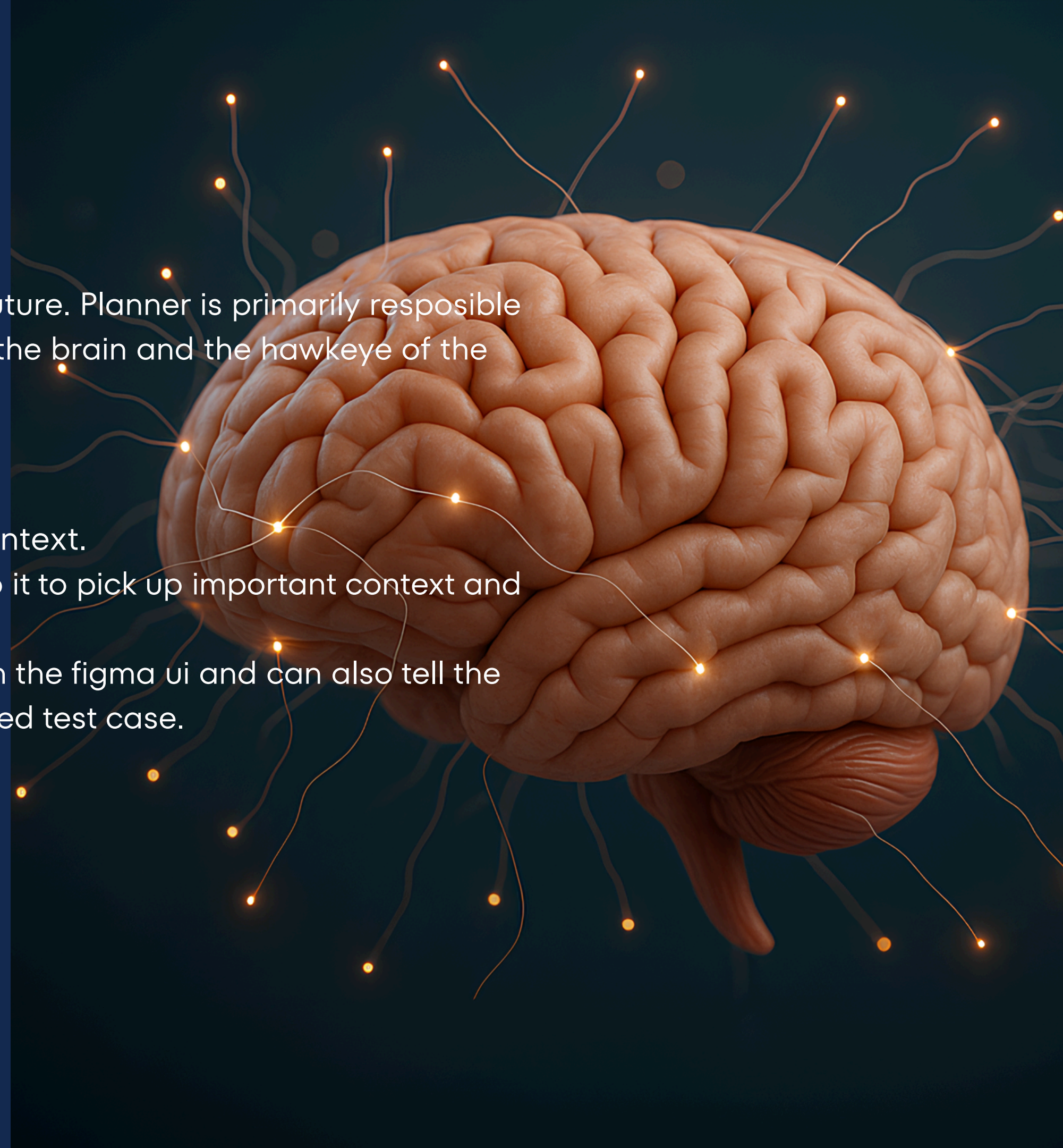
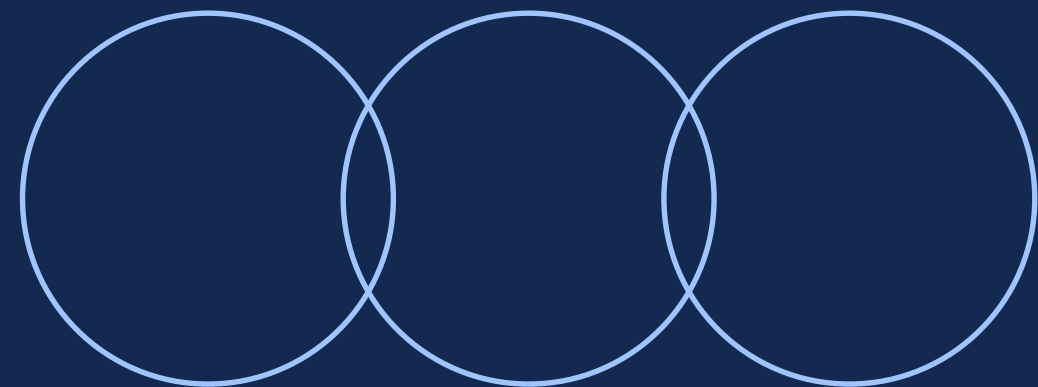


Planner Workflow

Planner is one of the core part of our overall architecture. Planner is primarily responsible for guiding which test case should be tested next. It is the brain and the hawkeye of the entire project.

What are we supposed to build?

- Direct the agent to give test cases with all the context.
- Help the agent navigate the entire context and help it to pick up important context and not bloating up the entire agent or wasting tokens
- Can give really intelligent advice to player based on the figma ui and can also tell the player what to verify for passing or failing the required test case.

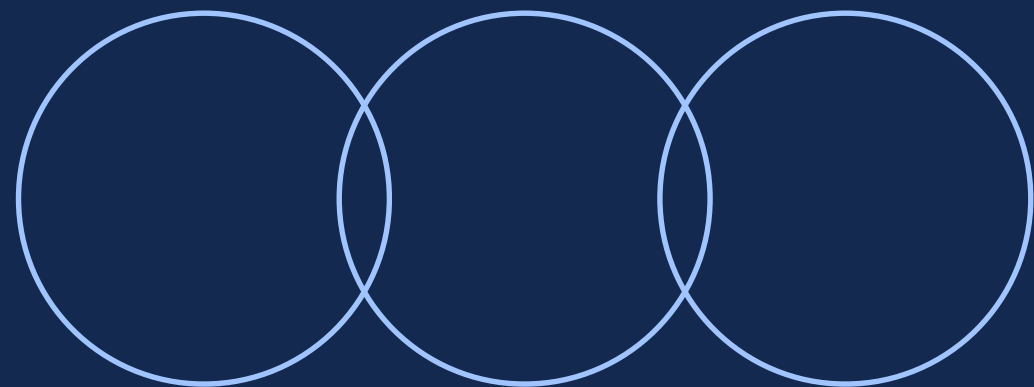


Player Workflow

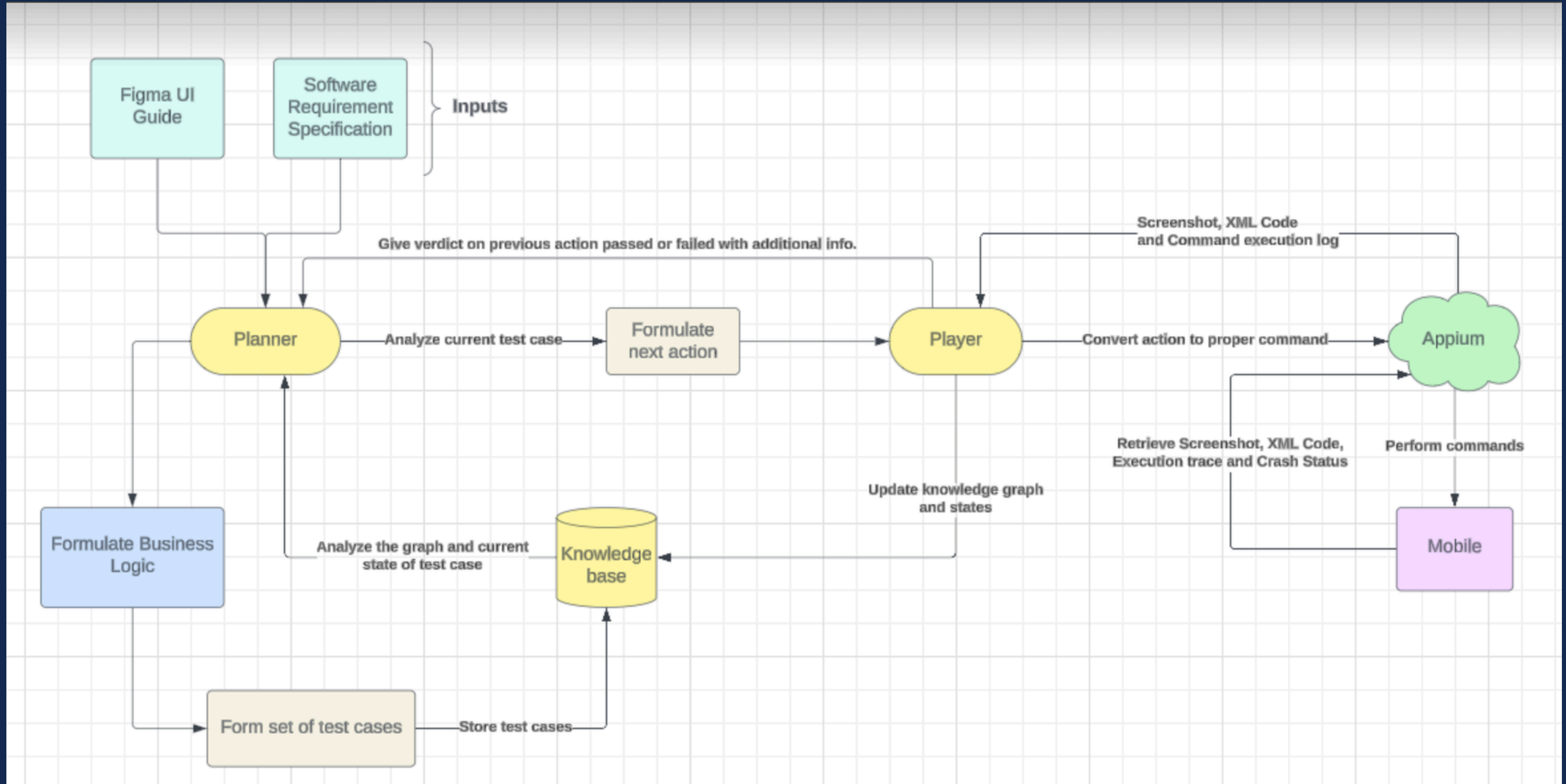
Player is probably the most essential part of the architecture which simulated the android device given the test case by the planner. It would later also build up with the other screens whose context are not available in the figma while exploring. The player needs to be intelligent enough to decide what to do and what not to.

What are we supposed to build?

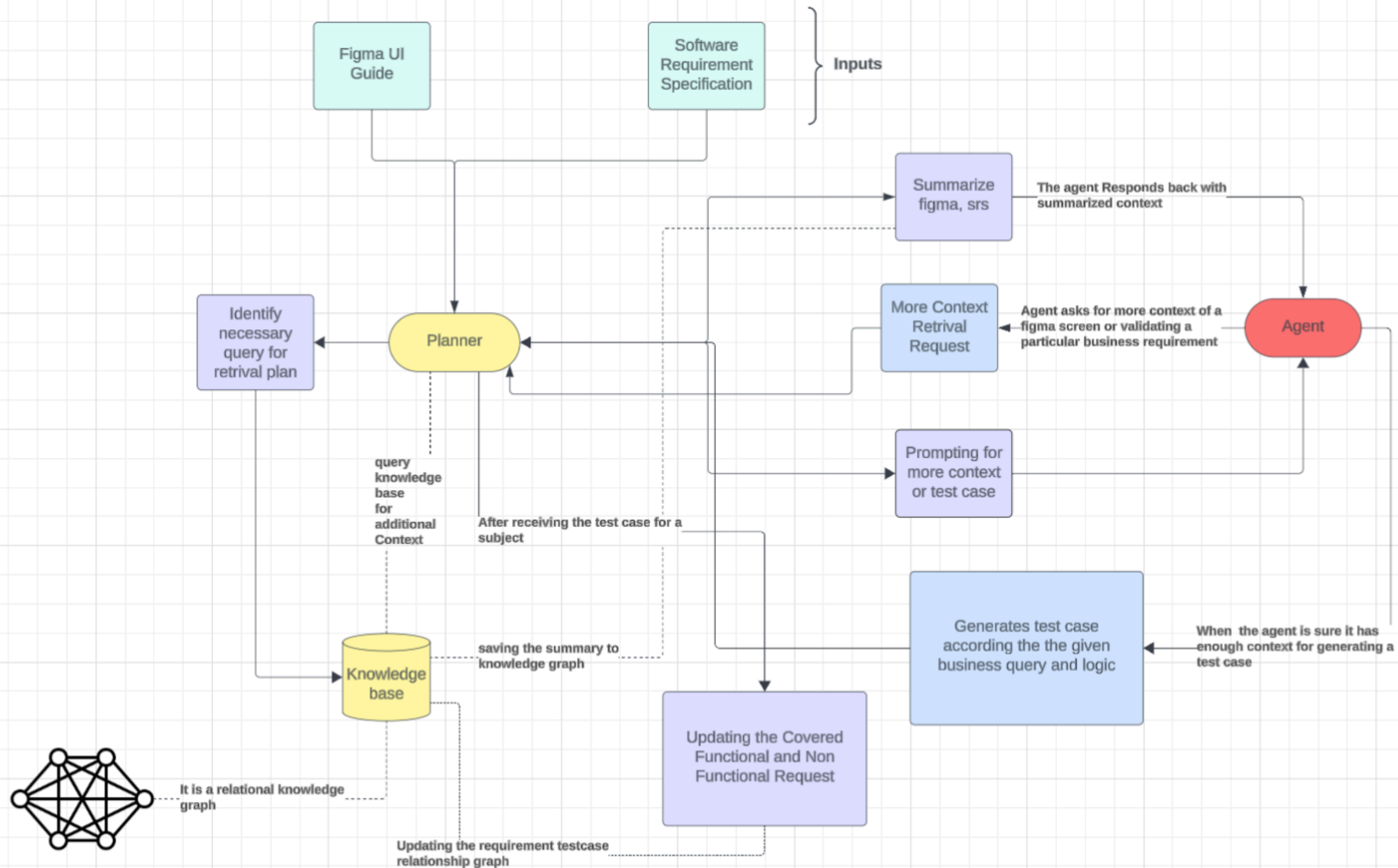
- Acts up the commands given by the planner.
- Tests if the given test case actually passes or fails and gives verdict.
- Needs a vlm and additional contexts to run and probably the most heaviest part of the entire system architecture..



Overall Architecture



Planner Agentic Architecture



Player Choice

As the player we are choosing Droid run
Because we have empirically tested it to be
one of the best and it has really impressive stats
on some of the agentic benchmarks

Mobilerun

[DOCS](#)[CLOUD](#)[Stars](#)

8.1k

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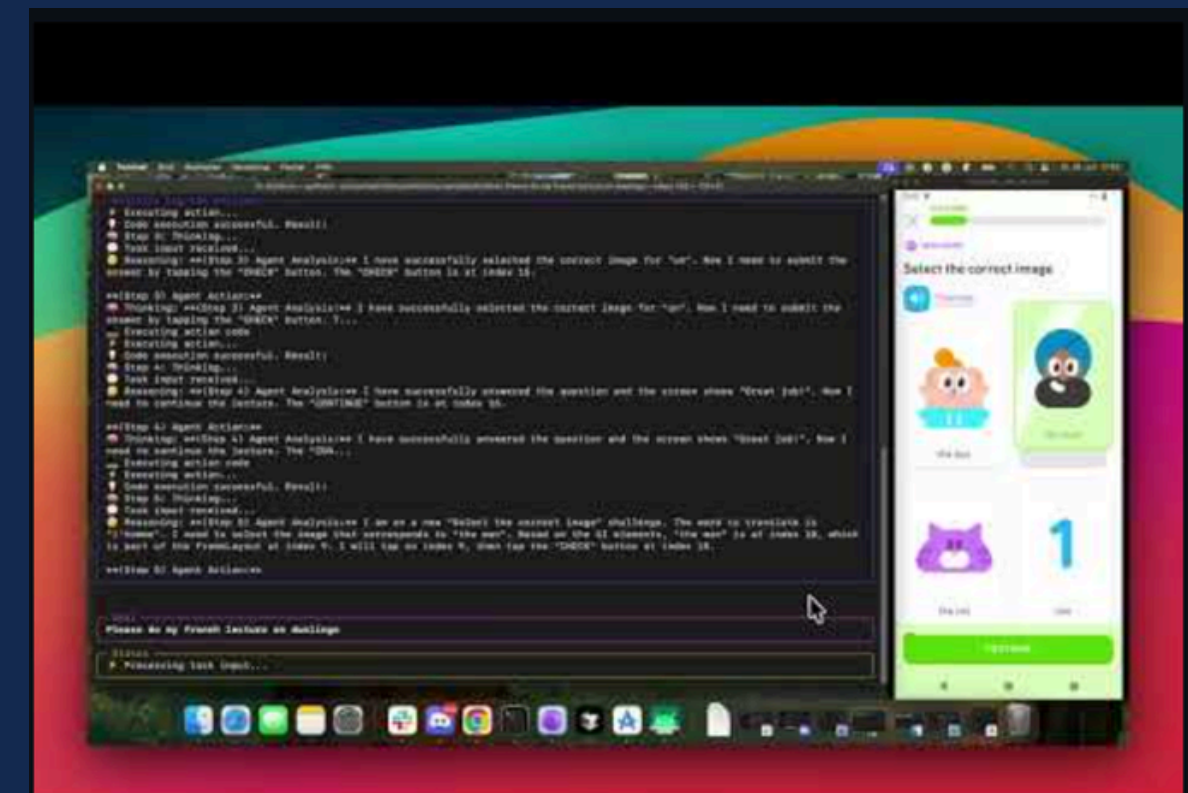
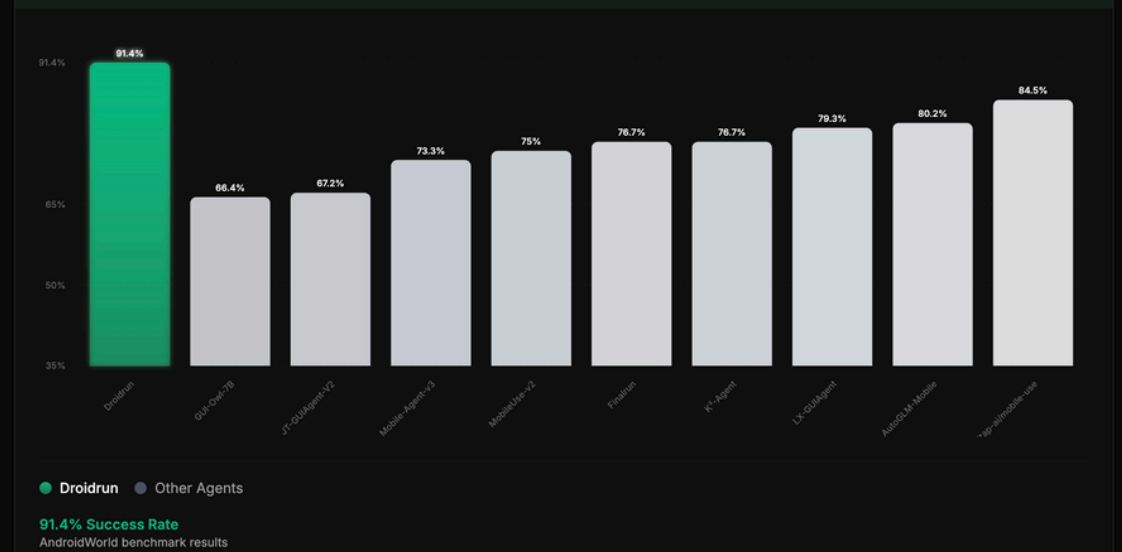
91 online

[Benchmark](#)

91.4%

AndroidWorld Benchmark Results

Success rates of leading AI agents on the 116-task AndroidWorld benchmark (03.10.2025)



Knowledge Base

Graph-first QA memory: Neo4j stores SRS, Figma UI, and execution history in one connected project graph.

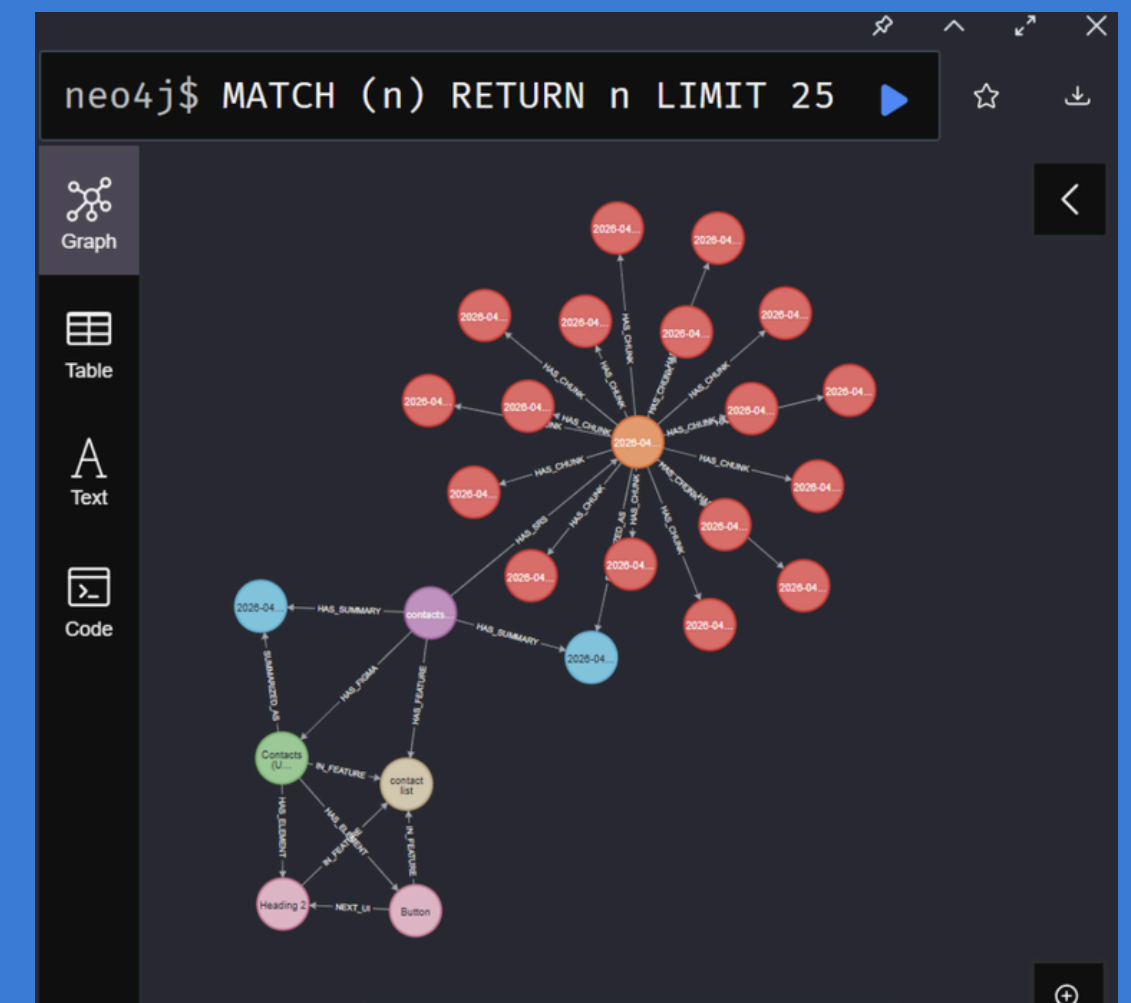
Requirement-level modeling: SRS is ingested into requirement-aware chunks (FR/NFR), enabling precise retrieval.

Coverage tracking: Each executed test is linked to covered requirements via COVERS_FR, so uncovered FRs are visible.

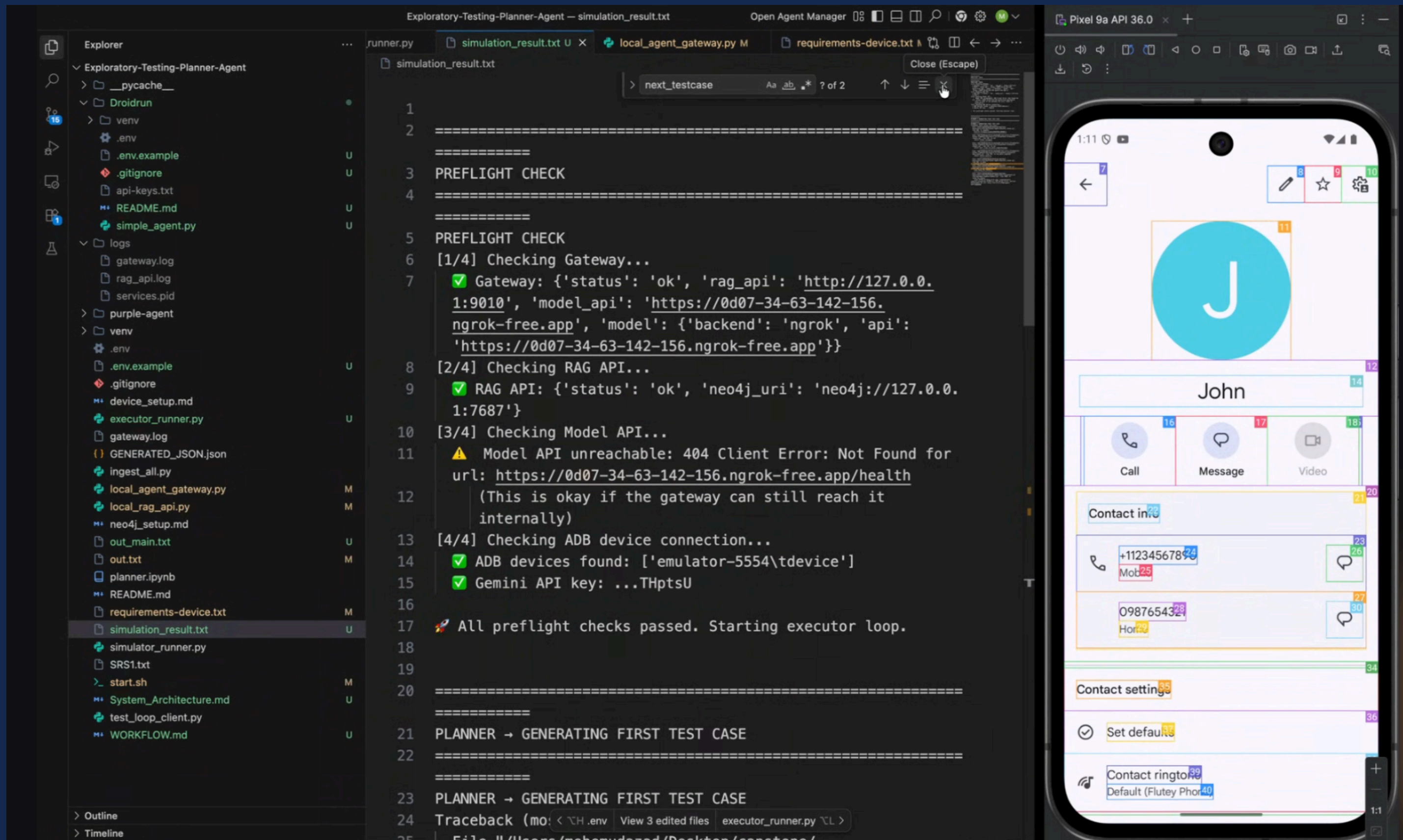
Context-optimized retrieval: Planner gets compact summaries first, then targeted multi-round retrieval to reduce token waste.

UI-aware test design: Figma screens/elements and inferred flow edges help generate realistic, element-referenced steps.

Observability: Debug traces expose planner decisions, retrieval context, and final generation prompts/outputs.



Live working interface



Example Prompt type

```
01 return (
02     f"You are retrieval planner for QA testcase generation in {app_name}.\n"
03     "You are interacting with a knowledge database (SRS + Figma KG + test history).\n"
04     "Decide your NEXT ACTION only.\n\n"
05     f"Objective: {objective}\n"
06     f"Retrieval round: {retrieval_round}/{max_rounds}\n"
07     f"Collected queries so far: {collected_queries}\n"
08     f"Collected screens so far: {collected_screens}\n"
09     f"Collected context size (chars): {context_chars}\n\n"
10     f"{response_link_line}"
11     "Retrieved context so far (continuation from earlier requests):\n"
12     f"{retrieved_notes_str}\n\n"
13     f"{global_context_block}"
14     "Return STRICT JSON only with this schema:\n"
15     "{\n"
16     "  \"action\": \"retrieve\" | \"produce_testcase\",\n"
17     "  \"retrieval_requests\": [{\"source\": \"srs|figma_ui|figma_flow\", \"query\": \"...\"},\n"
18     "  \"focus_queries\": [\"...\", \"...\"],\n"
19     "  \"target_screens\": [\"...\", \"...\"],\n"
20     "  \"reason\": \"short reason\"\n"
21     "}\n"
22     "Rules:\n"
23     "- If more context is needed, set action=retrieve and provide explicit retrieval_request.\n"
24     "- Use source=srs for business rules/validation logic.\n"
25     "- Use source=figma_ui for screen elements and control availability.\n"
26     "- Use source=figma_flow for navigation/button-to-screen behavior.\n"
27     "- For source=srs, query must be keyword expression (not question), e.g. \"validation AND\n"
28     "- You are replying to the prior retrieval results; avoid asking for the exact same request.\n"
29     "- If context is sufficient, set action=produce_testcase.\n"
30     "- Never output markdown or text outside JSON."
31 )
```

Retreival Request

```
[Planner round raw answer]:
{
  "action": "retrieve",
  "retrieval_requests": [
    {
      "source": "srs",
      "query": "validation AND (email OR phone)"
    },
    {
      "source": "figma_ui",
      "query": "Create Contact screen elements",
      "screen": "create_contact"
    },
    {
      "source": "figma_flow",
      "query": "navigation from Create Contact to Contacts List"
    }
  ],
  "focus_queries": [
    "validation rules for contact creation",
    "UI elements available for input on Create Contact screen"
  ],
  "target_screens": [
    "create_contact",
    "contact_list"
  ],
  "reason": "Need to retrieve validation rules for contact creation and UI elements on the Create Contact screen to generate a relevant and non-duplicate test case."
}
```

Test Case Generation

```
```json
{
 "test_case_id": "TC-012",
 "title": "Verify system displays fallback identifiers in contact list when name is missing",
 "screen": "Contacts List (Updated)",
 "preconditions": [
 "A contact is created without a name but with at least one valid phone number or email address",
 "The contact is saved to the local storage"
],
 "steps": [
 "Open the Contacts app",
 "Navigate to the 'Contacts List (Updated)' screen",
 "Locate the contact that was saved without a name",
 "Verify that the contact list displays the phone number or email as the fallback identifier"
],
 "expected_result": "The contact list displays the phone number or email instead of a name for contacts without a name.",
 "priority": "high",
 "area": "UI Behavior",
 "rationale": "This test directly addresses a previously failed test and validates FR-11 (fallback identifiers in the list). It ensures the system adheres to the SRS requirement and improves UI reliability."
}
```
```

Successful Bug Finding

Presets

ET Executor

Search for logs

Last 2 days

Apr 7 at 1:39:11.981 GMT+06

| Time | Source | Level | Message |
|--------------------------------|-------------|-------|---|
| 2026-04-07 01:37:45.260 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
PLANNER → LOGGING VERDICT & G |
| 2026-04-07 01:38:31.055 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
Planner RAG interaction [roun |
| 2026-04-07 01:38:31.056 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
Verdict logged: TC-012 → pass |
| 2026-04-07 01:38:31.057 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
ROUND 2/2 |
| 2026-04-07 01:38:31.058 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
EXECUTING ON DEVICE: TC-013 |
| 2026-04-07 01:38:31.058 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
Droidrun goal dispatched for |
| 2026-04-07 01:39:11.981 GMT+06 | ET Executor | WARN | context.system.pid:69060 cor
Test TC-013 execution: FAILED |
| 2026-04-07 01:39:11.982 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
ROUND 2 RESULT |
| 2026-04-07 01:39:11.982 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
LOGGING FINAL VERDICT |
| 2026-04-07 01:39:12.090 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
EXECUTION SUMMARY |
| 2026-04-07 01:39:12.090 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
Execution complete: 1/2 passe |
| 2026-04-07 01:39:12.132 GMT+06 | ET Executor | INFO | context.system.pid:69060 cor
RECENT TESTS IN NEO4J |

Log Apr 7, 2026 at 1:39:11.981am GMT+06

ET Executor • WARNING context.system.pid:69060 context.runtime.thread_id:8739364992 context.runtime.file:executor_runner.py context.runtime.line:72

Overview Context Explain with AI Comments

{ } JSON

Expand all

dt: "2026-04-06T19:39:11.981399+00:00", // Apr 7, 2026 at 1:39:11am GMT+06

level: "warning",

severity: 3, // TRACE3

message: "Test TC-013 execution: FAILED",

context: {

runtime: {

function: "cloud_log",

file: "executor_runner.py",

line: 72,

thread_id: 8739364992,

thread_name: "MainThread",

logger_name: "executor"

}

system: {

pid: 69060,

process_name: "MainProcess"

}

filename: "executor_runner.py",

stack_info: null,

taskName: "Task-1",

test_case_id: "TC-013",

title: "Verify the system allows the user to save a contact with a valid phone number and an invalid email address and prevents saving due to FR-5 and FR-7",

verdict: "failed",

steps_taken: 6,

duration_seconds: 40.9,

reason: "The contact was saved with an invalid email address. No error message was displayed. The expected result was not achieved."

}



Contributions

Mohammad Raihan Rashid: Overall Planner Design and Base Version Creation, Figma and Srs ingestion pipeline, Presentation, Overall Architecture Design and literature review.

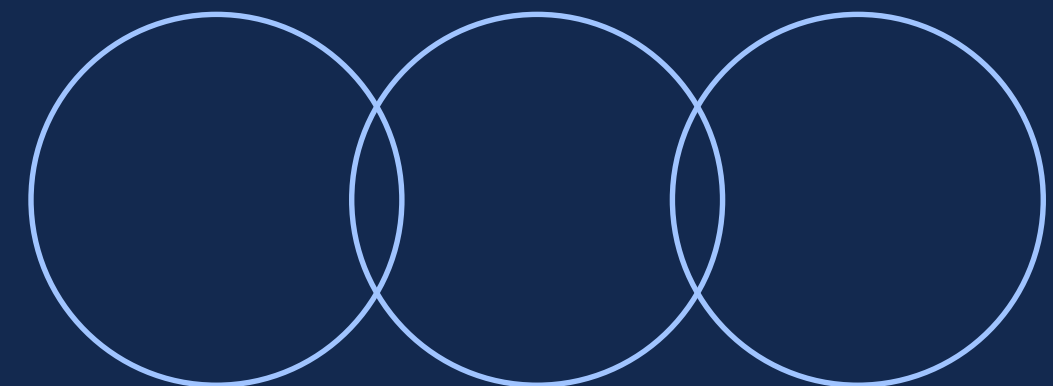
Mehemud Azad: Integrating the current planner with Droid run Player, Extensive player exploration, Making a ui for planner, player overall test results. Bug fixes in the planner and player configuration changes.

Jonayed Mohiuddin Shanto: Figma (used mcp server to get the json context) and Srs Creation. Extensive Literature review. Integrating App Agent X with the current planner. Main Architecture Design Contribution.

Khalid Hasan Tuhin: Atomize the function requirements, showing the communication through a UI, Investigate the correctness of figma graph and ingestions.

Niloy Kumar Mondal: Visualization of Graph structure of the knowledge base, Some literature review.

Github Link to our project: <https://github.com/raihanrsdV1/Exploratory-Testing-Planner-Agent>



Any questions or thoughts?

We appreciate your attention and participation today

